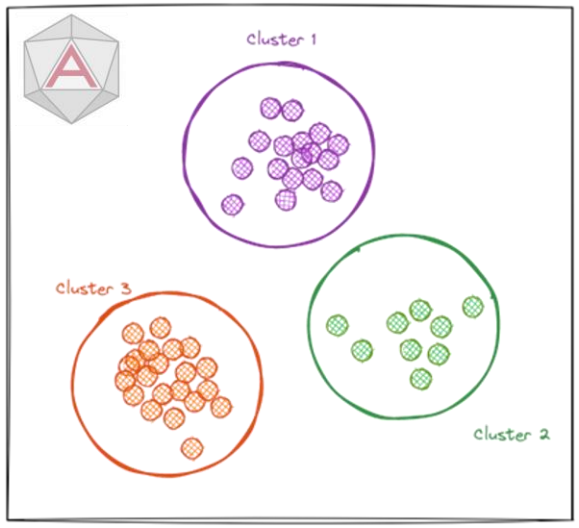
ITEC 4220 Adv Data Analytics (Modified from CIS4930: Data Science Foundations)

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Cluster Analysis

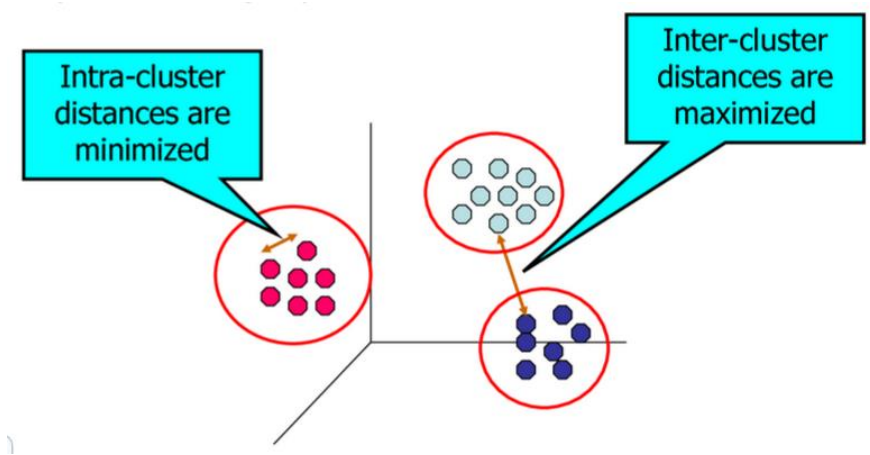
Clustering: The Main Idea

- "Unsupervised" learning technique.
- Goal: Form groups (clusters) of similar records.
- Segment data points into homogeneous and meaningful clusters.



Clustering: The Main Idea

- Desired properties of clustering result:
  - ✓High intra-cluster similarity, low inter-cluster similarity.



Clustering

- Helps to gain insights into your data.
- Applications:
  - ✓Market segmentation,
  - ✓Medical diagnostics,
  - ✓Bioinformatics,
  - ✓Text mining/ information retrieval, etc.

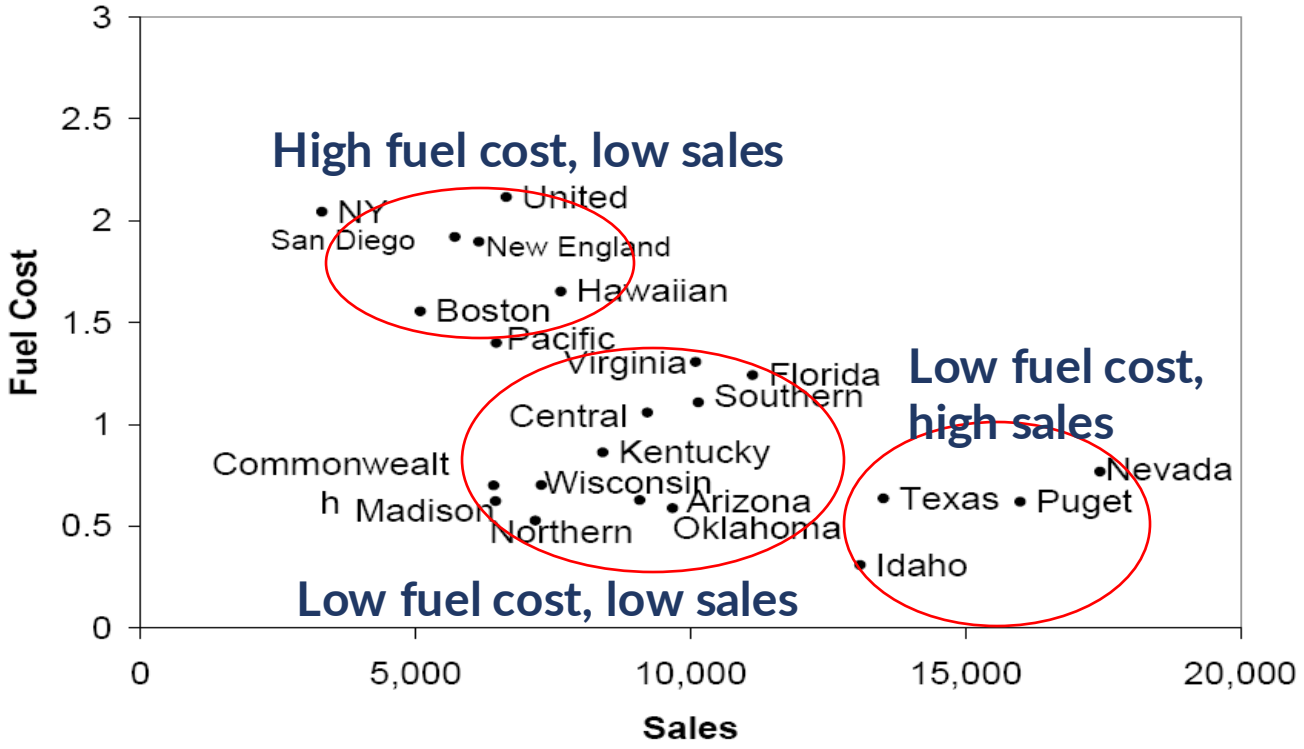
Example: Public Utilities

- **Goal:** Based on the information about different public utility companies, **find clusters/groups of similar utilities**
- **Data:** 22 firms, 8 variables
  - ✓Fixed-charge covering ratio
  - ✓Rate of return on capital
  - ✓Cost per kilowatt capacity
  - ✓Annual load factor
  - ✓Growth in peak demand
  - ✓Sales
  - ✓% nuclear
  - ✓Fuel costs per kwh

Company	Fixed_charge	RoR	Cost	Load	Δ Demand	Sales	Nuclear	Fuel_Cost
Arizona	1.06	9.2	151	54.4	1.6	9077	0	0.628
Boston	0.89	10.3	202	57.9	2.2	5088	25.3	1.555
Central	1.43	15.4	113	53	3.4	9212	0	1.058
Commonwealth	1.02	11.2	168	56	0.3	6423	34.3	0.7
Con Ed NY	1.49	8.8	192	51.2	1	3300	15.6	2.044
Florida	1.32	13.5	111	60	-2.2	11127	22.5	1.241
Hawaiian	1.22	12.2	175	67.6	2.2	7642	0	1.652
Idaho	1.1	9.2	245	57	3.3	13082	0	0.309
Kentucky	1.34	13	168	60.4	7.2	8406	0	0.862
Madison	1.12	12.4	197	53	2.7	6455	39.2	0.623
Nevada	0.75	7.5	173	51.5	6.5	17441	0	0.768
New England	1.13	10.9	178	62	3.7	6154	0	1.897
Northern	1.15	12.7	199	53.7	6.4	7179	50.2	0.527
Oklahoma	1.09	12	96	49.8	1.4	9673	0	0.588
Pacific	0.96	7.6	164	62.2	-0.1	6468	0.9	1.4
Puget	1.16	9.9	252	56	9.2	15991	0	0.62
San Diego	0.76	6.4	136	61.9	9	5714	8.3	1.92
Southern	1.05	12.6	150	56.7	2.7	10140	0	1.108
Texas	1.16	11.7	104	54	-2.1	13507	0	0.636
Wisconsin	1.2	11.8	148	59.9	3.5	7287	41.1	0.702
United	1.04	8.6	204	61	3.5	6650	0	2.116
Virginia	1.07	9.3	174	54.3	5.9	10093	26.6	1.306

Example

- For example, if we consider only sales & fuel cost:
- 3 rough clusters can be seen



Things to Think About

- Similarity metrics are important for clustering
  - ✓Maximize intra-cluster similarity (“tight” cluster)
  - ✓Minimize inter-cluster similarity (“distinct” cluster)
  - ✓But what kind of similarity to use?
- Stopping criteria
  - ✓How many clusters should you have?
- Clustering algorithms and their parameters
  - ✓How to come up the clusters?

Distance Metric

- A key component of most clustering techniques is the **distance metric.**
- Therefore, the choice of a metric is important
  - ✓Distance between two points.
  - ✓Distance between clusters.

Distance Measurement

- **Euclidean distance** is the most popular measurement

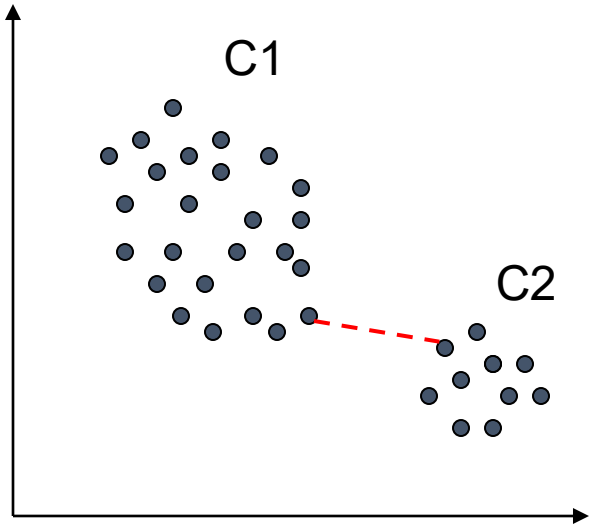
$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}$$

- It is highly influenced by the scale of each variable:
  - ✓Scale the variables

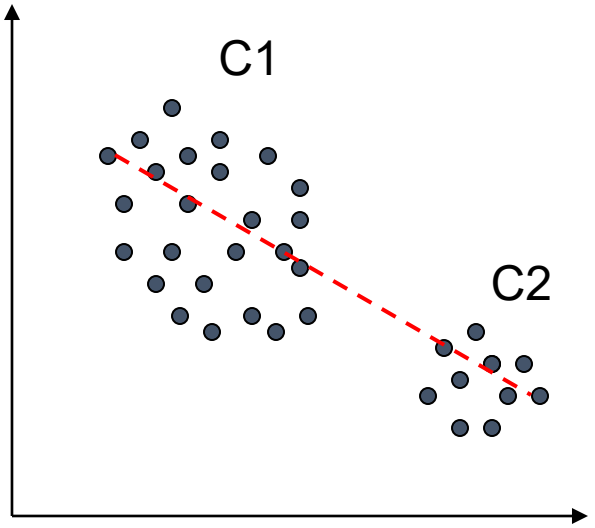
Distance Between Clusters

- The most widely used measures are:
  - ✓Minimum distance (distance between the two closest points in each cluster)
    - Use when you care about connected structure or unusual shapes
  - ✓Maximum distance (distance between the two farthest points in each cluster)
    - Use when you care about compact, well-separated clusters
  - ✓Average distance (the average distance of all possible distances between records of each cluster)
    - Use when you want a balance between the two.

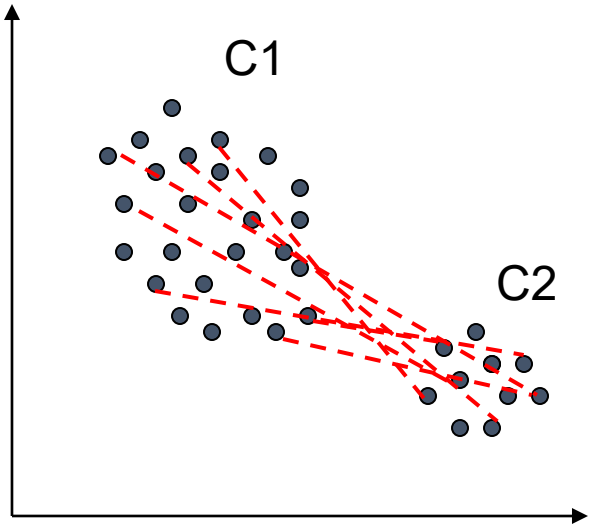
Distance Between Clusters



Single linkage
(minimum pairwise distance between points from two different clusters)



Complete linkage
(maximum pairwise distance between points from two different clusters)



Average linkage
(average pairwise distance between points from two different clusters)

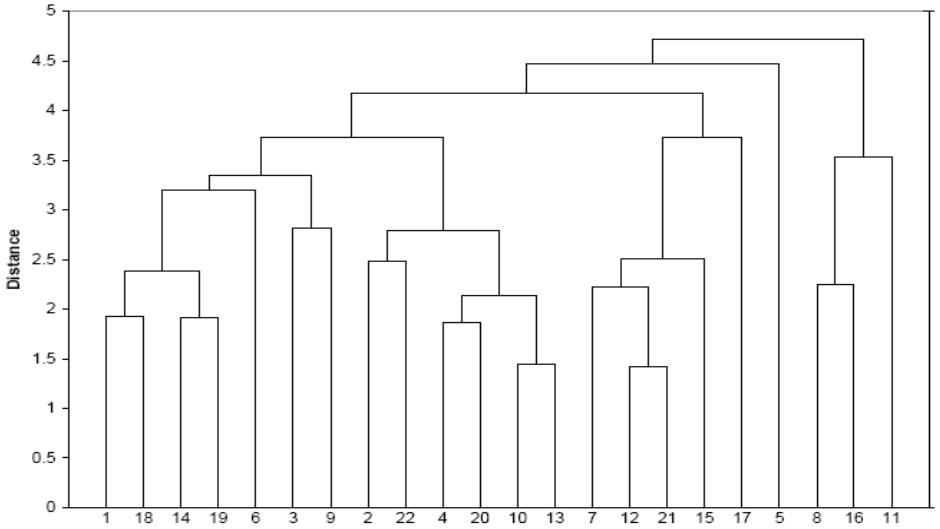
Two Popular Clustering Approaches

- Hierarchical clustering:
  - ✓Observations are sequentially grouped to create clusters.
  - ✓Based on distances between observations and distances between clusters.
- K-means clustering (Nonhierarchical method):
  - ✓Observations are allocated to a pre-specified set of clusters.
  - ✓Based on their distance from each cluster.
  - ✓Less computationally intensive and preferred for very large datasets.

Hierarchical Clustering

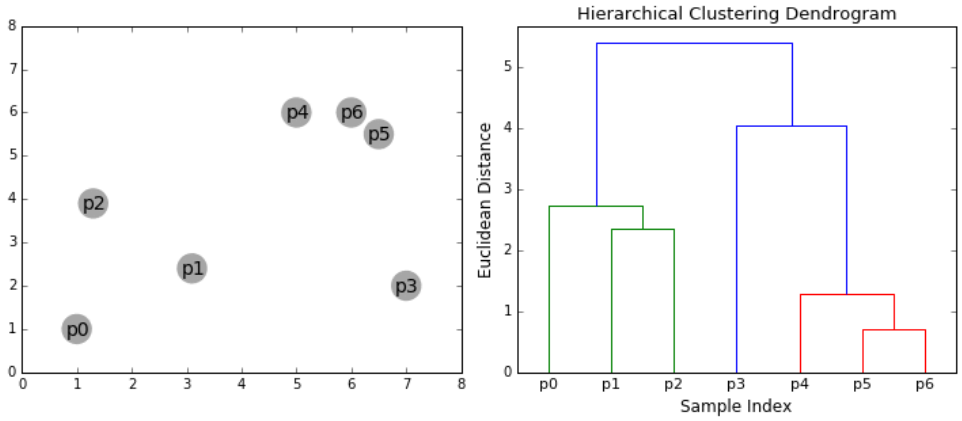
Hierarchical Methods

- Organizes data into a tree-like structure, where smaller groups of similar data points are progressively merged into larger clusters.



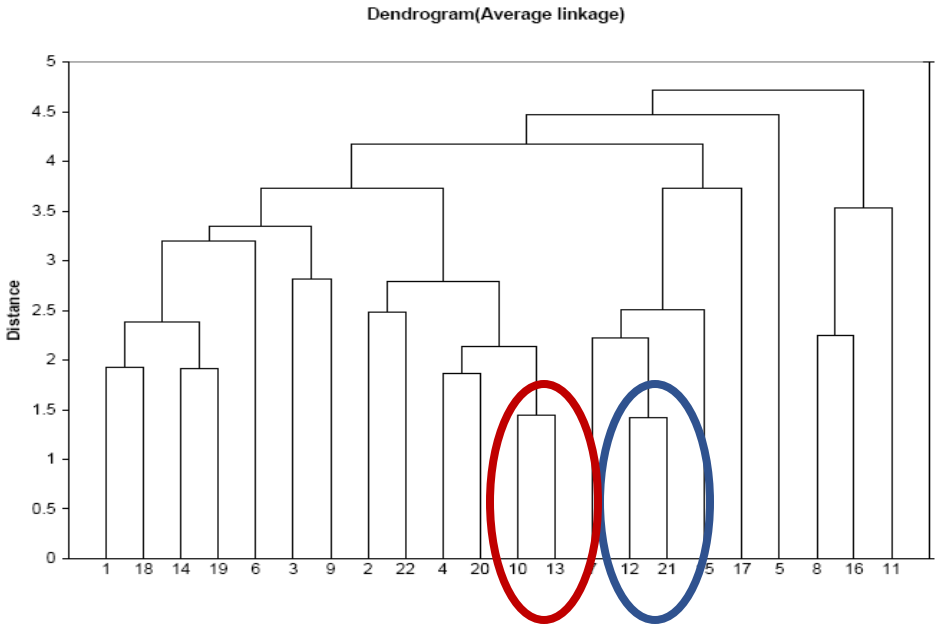
Hierarchical Methods

- **Start with each point as its own cluster**: Initially, every data point is considered as its own individual cluster.
- **Merge the closest clusters**: At each step, the two clusters that are closest to each other (based distance metric) are merged into one larger cluster.
- **Repeat until one large cluster remains**: This process continues until all data points are merged into one single cluster, or until a predefined number of clusters is reached.



Reading the Dendrogram

- **See process of clustering:** Lines connected lower down are merged earlier:
  - ✓12 & 21 merged first!
  - ✓10 and 13 merged next!



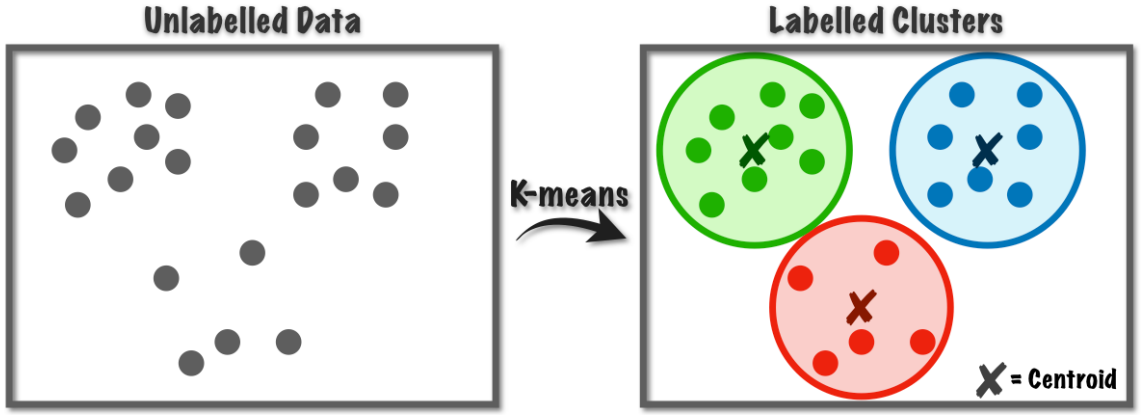
Hierarchical Clustering

- Advantages
  - ✓Does not require prespecify number of clusters.
  - ✓Dendrogram represents the cluster process and results in an intuitive way.
- Limitations
  - ✓Slow & computational expensive
    - Need to compute *n* x *n* distance matrix
  - ✓Sensitive to outliers

K-Means Clustering

K-Means Clustering Algorithm

- Number of clusters, K, must be specified.
- Each cluster is associated with a centroid (center point).
- Each point is assigned to the cluster with the closest centroid.
- The objective is to minimize the sum of distances of the points to their respective centroid.



K-Means Clustering Algorithm

1. Prespecify the desired # of clusters, *k*.
2. Start with a partition into k clusters.

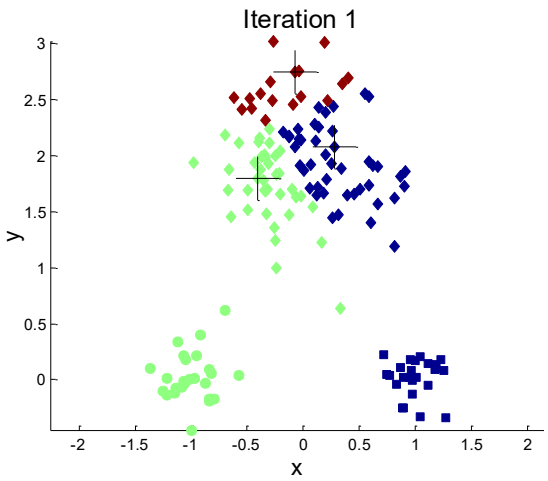
Often based on random selection of k centroids

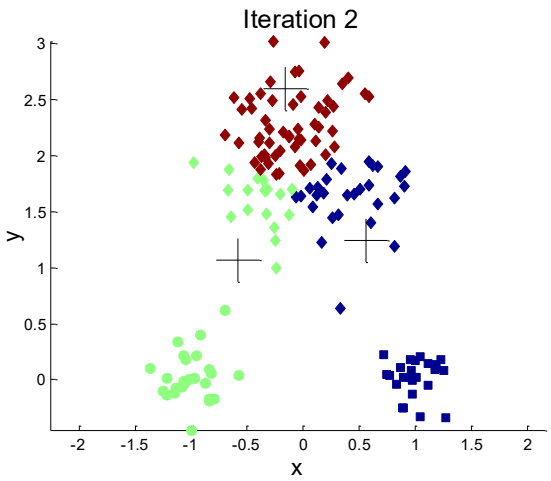
3. At each step, move each record to cluster with closest centroid.
4. Recompute centroids, repeat step 3.
5. Stop when moving records increases within-cluster dispersion.

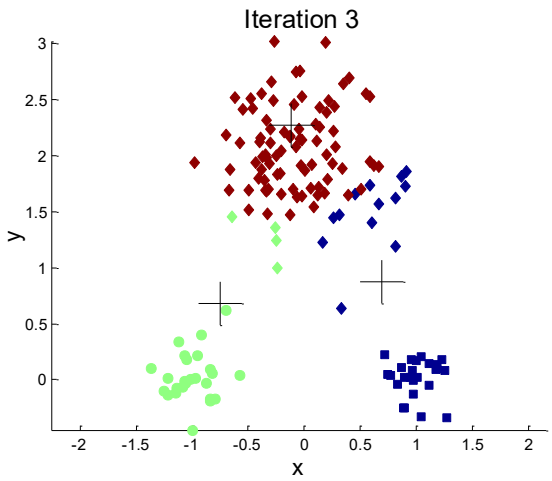
How Do We Find the Centroid?

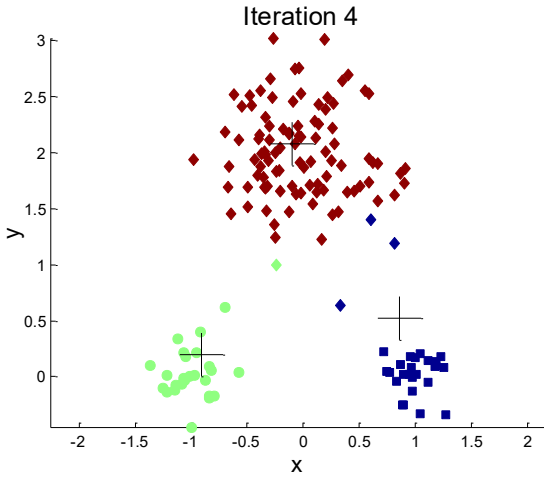
- Initial centroids are often chosen randomly.
- Clusters produced vary from one run to another.
- Do multiple runs and select the clustering with the smallest error.

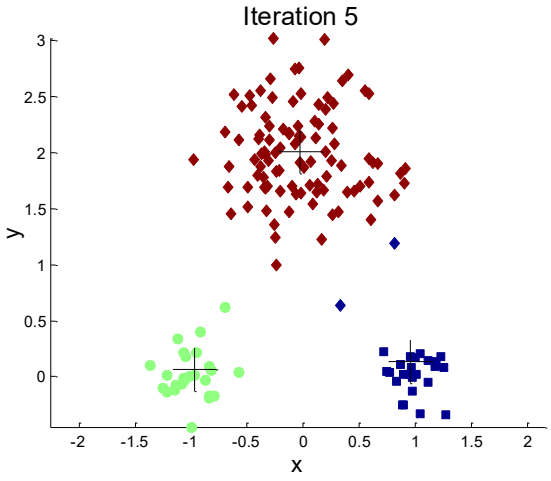
The Importance of Initial Centroids

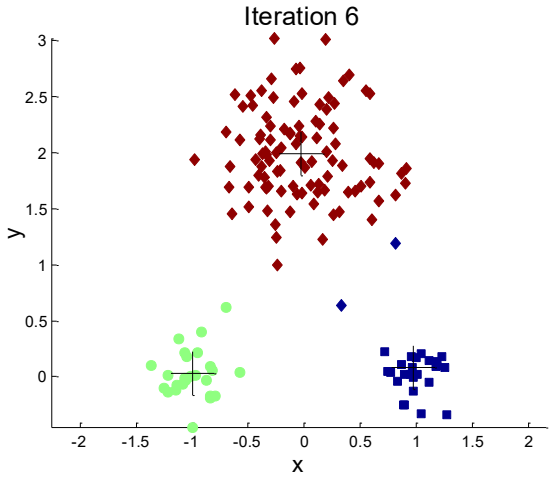




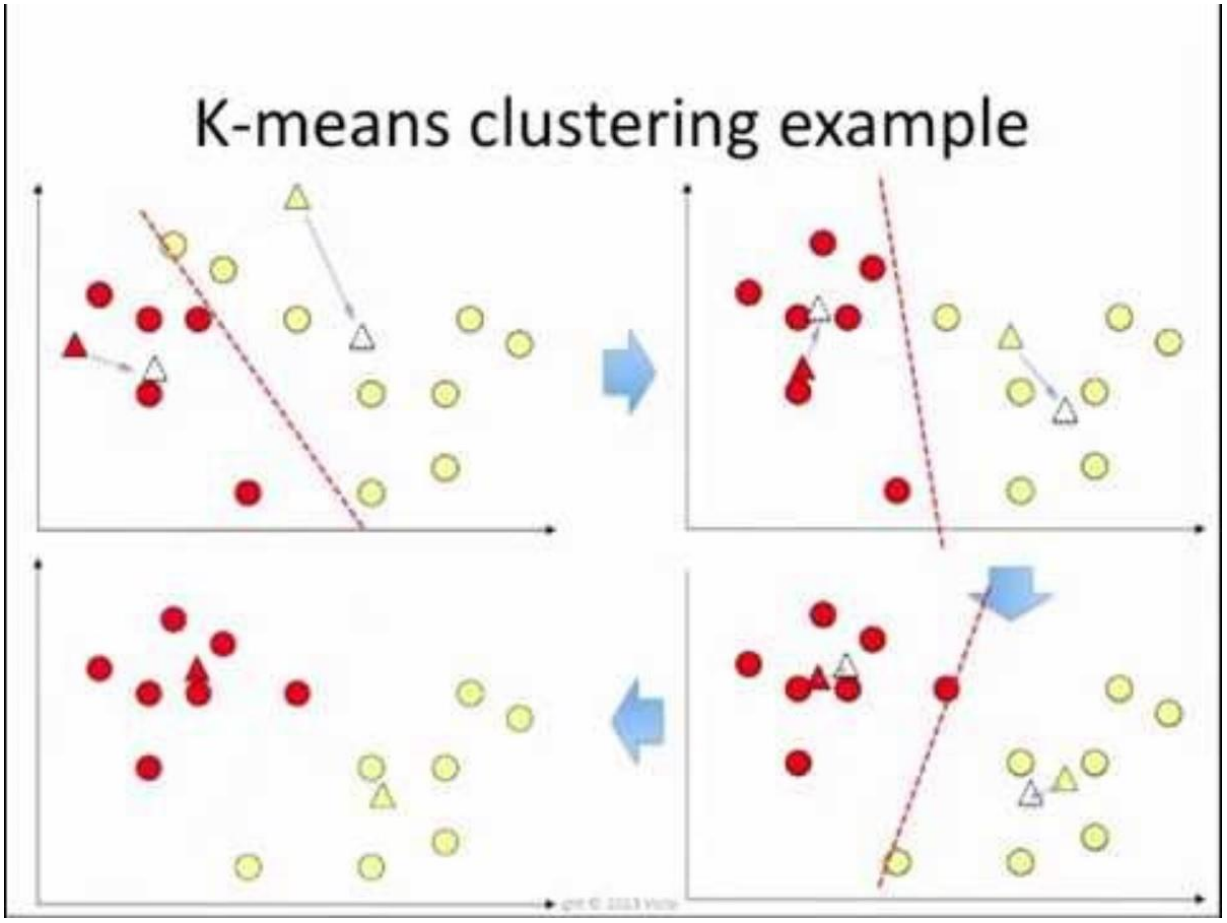








K-means Demo



Pre-process to Get Good Centroids

- Get the data ready for analysis
  - ✓Deal with missing values
- Scale the Data
- Remove Outliers

Evaluating K-means: Inertia

- Inertia measures how far data points are from their assigned cluster centers.
- Intuitively, it tells us how compact the clusters are.
- Lower inertia means tighter, more coherent clusters.

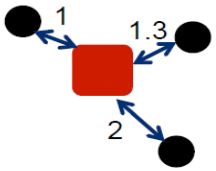
Within Cluster Inertia

- For one cluster C_k with centroid μ_k:

$$\mathrm{WCSS}_k = \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

xi = data point in cluster k
μ_k =centroid of cluster k

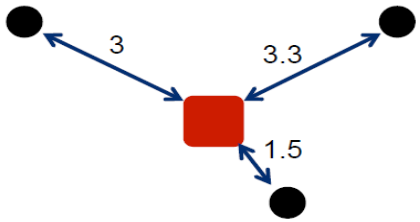
Cluster 1



$SSE_1 = 1^2 + 1.3^2 + 2^2$

$= 1 + 1.69 + 4 = 6.69$

Cluster 2



$SSE_2 = 3^2 + 3.3^2 + 1.5^2$

$= 9 + 10.89 + 2.25 = 22.14$

Total Inertia

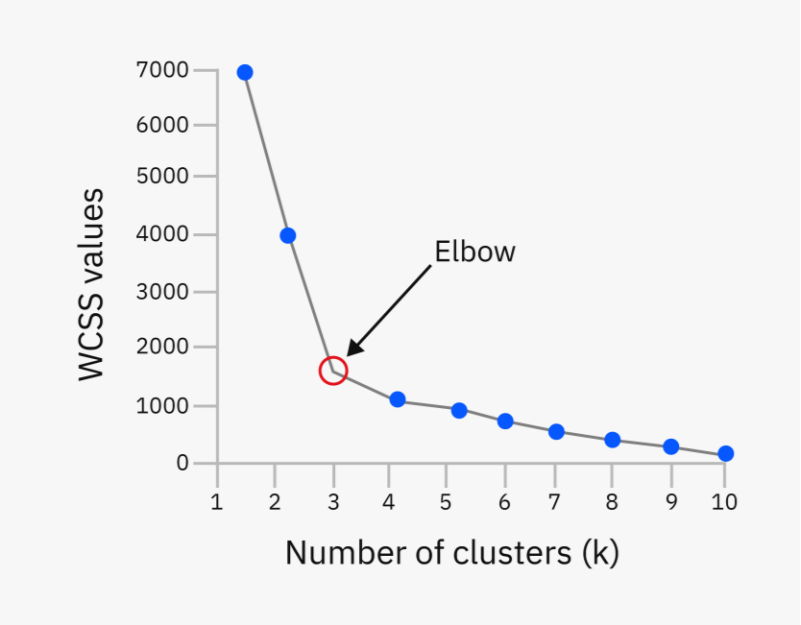
- Total inertia is the sum of all within-cluster inertias

$$\text{Inertia} = \sum_{k=1}^{K} \text{WCSS}_k$$

- We use total inertia to find the optimal k using a method called **elbow method**

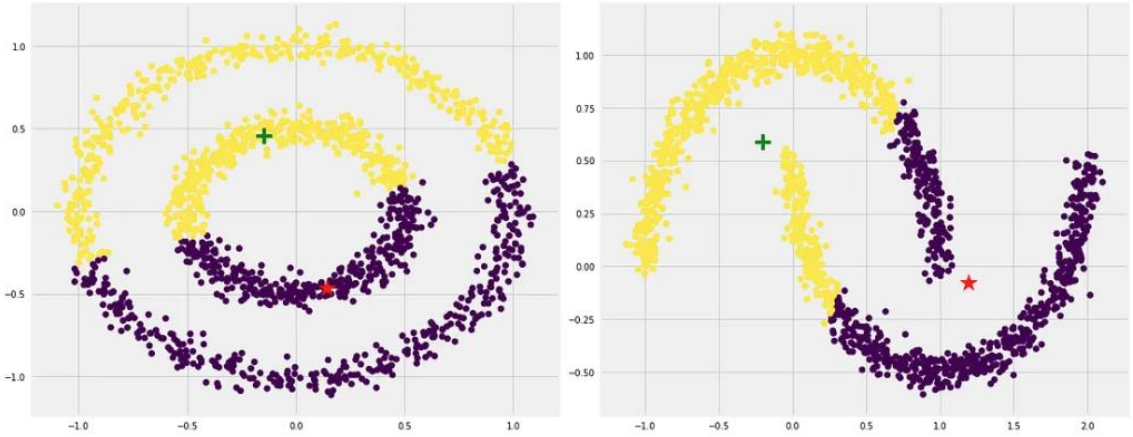
Elbow Method

- In the graph, the point we observe elbow shows the optimal number of k.
- We see significant decrease in inertia until the elbow point
- After that point, the decrease is not very significant.



Limitations of k-Means Clustering

- K-Means gives unreliable results when…
  - ✓Clusters vary widely in size
  - ✓Clusters vary widely in density
  - ✓Clusters are not in rounded shapes



When to Use Which?

- Use Hierarchical:
  - ✓ Small dataset
  - ✓ You do not know k
  - ✓ Flexibility in cluster shapes
  - ✓ Exploratory analysis
- Use K-means:
  - ✓ Larger dataset
  - ✓ Clusters are roughly spherical
  - ✓ You know (or can guess) k
  - ✓ Goal is partitioning